

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS – SSC-I (9th Grade)

Total Marks: 65

Time Allowed: 2 hours 30 minutes

Version: 2

General Instructions:

1. This paper has **three sections**: A (MCQs), B (Short Questions), C (Long Questions).
2. **Section A** (12 marks): 12 MCQs. Mark the correct bubble. **Section B** (33 marks): 11 short questions × 3 marks each.
3. **Section C** (20 marks): 4 long questions × 5 marks each. Answer **either** the main question **or** the OR alternative in B and C.
4. Scientific calculator allowed. Blue/black ink pen only. Write legibly.

SECTION A – Multiple Choice Questions (12 × 1 = 12 Marks) [15 min]

Q.1: Choose the correct answer and fill in the corresponding bubble.

#	Question	A	B	C	D	Answer
1	Which is a non-contact force?	Friction	Tension	Gravitational force	Normal force	☐ A ☐ B ☒ C ☐ D
2	Newton's first law is also called the law of:	Acceleration	Inertia	Action-reaction	Momentum	☐ A ☒ B ☐ C ☐ D
3	The SI unit of force ($F = ma$) is:	kg·m/s	kg·m/s ²	N·s	kg/m ²	☐ A ☒ B ☐ C ☐ D
4	Action and reaction forces act on:	Same body	Different bodies	Same point	Same direction	☐ A ☒ B ☐ C ☐ D
5	The SI unit of momentum ($p = mv$) is:	N·m	kg·m/s ²	kg·m/s	J/s	☐ A ☐ B ☒ C ☐ D
6	The weakest fundamental force is:	Strong nuclear	Electromagnetic	Weak nuclear	Gravitational	☐ A ☐ B ☒ C ☐ D
7	Torque is the product of force and:	Mass	Velocity	Moment arm	Time	☐ A ☐ B ☒ C ☐ D
8	For complete equilibrium, both must be zero:	Force & mass	Net force & net torque	Velocity & acceleration	Momentum & energy	☐ A ☒ B ☐ C ☐ D
9	A cone resting on its base is in:	Stable equilibrium	Unstable equilibrium	Neutral equilibrium	Dynamic equilibrium	☒ A ☐ B ☐ C ☐ D
10	Which does not reduce friction?	Polishing	Ball bearings	Lubricants	Increasing roughness	☐ A ☐ B ☒ C ☐ D
11	If speed is doubled in $F_c = mv^2/r$, force becomes:	Double	Half	Four times	Same	☐ A ☐ B ☒ C ☐ D
12	A tentative assumption that can be verified is a:	Law	Theory	Hypothesis	Principle	☐ A ☐ B ☒ C ☐ D

Candidate's Signature

Invigilator's
Signature

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Section - B (Second paper)

_____ Q:2(i) _____
 _____ Q:2(1) _____

FORCE:-

Force is the vector quantity which changes or tends to change the state of body; starts or stops its motion, speeds it up or slows it down and can change the direction of its motion.

Contact forces**Non-contact forces****Definition**

- | | |
|--|---|
| <p>* The forces acting between two objects that are in physical contact are termed as contact forces</p> | <p>* The forces which act at a distance, without any physical contact between bodies are called non-contact forces.</p> |
|--|---|

Examples:-

- | | |
|---|--|
| <p>* In cricket, a batter hitting a cricket ball is a contact force between since there is a physical ^{contact} force in between the bat and ball.</p> | <p>* Gravitational force between Earth and the moon acts across the vacuum of space without any contact.</p> |
|---|--|

Q:2(ii)
Q:2(ii)

* Newton's first law of motion:-

"If the net external force acting on an object is zero, ~~then~~ the object will maintain its state of rest or uniform motion (constant velocity)."

Example:-

An object can continue to move when the unbalanced forces are removed. For example, when a soccer ball is kicked, it experiences an unbalanced force. The ball keeps rolling on the ground until another unbalanced force ~~alter~~ alters its movement.

✓ Inertia:-

Natural tendency of an object to remain its state of rest or in uniform in a straight line is called inertia.

Q: 2 (iii)
Q: 2 (iii)

* Newton's second Law of motion:-

"The acceleration of an object is as produced by the net force is directly proportional to the magnitude of the net force in the direction as the net force. in the same direction as the net force and inversely proportional to the mass of the object"

Mathematically,

$$F_{\text{net}} = ma$$

* Data:-

$$F = 3000 \text{ N}$$

$$\text{mass} = 12,000 \text{ kg}$$

Given:-

$$F = ma$$

$$\Rightarrow 3000 \text{ N} = 12,000 (a)$$

$$\Rightarrow \frac{3000}{12000} = a$$

$$\Rightarrow \frac{3}{12} = a$$

$$\Rightarrow 0.25 \text{ m/s} = a$$

$$\frac{0.25}{0.25} = \frac{0.25}{0.25} \quad \frac{0.25}{0.25} = \frac{0.25}{0.25}$$

① Momentum:-

The product of mass (m) and velocity ($\vec{v}$) is called momentum, denoted by ' $\vec{p}$ '. Mathematically,

$$\vec{p} = m\vec{v}$$

Momentum is a vector quantity that also has magnitude and direction. SI unit of momentum is kilogram-meter per second (kg m/s) or Newton-second (Ns). Newton's second law is used to relate force and momentum. $\therefore \text{force} = \frac{\text{change in momentum}}{\text{time}}$

② Data:-

mass = 45 g

velocity = 80 m/s

formula of momentum = $\vec{p} = m\vec{v}$

So, we will convert g into kg

$$\frac{45}{1000} = 0.045 \text{ kg}$$

$$\text{then } \vec{p} = \cancel{0.045 \text{ kg}} \times 80 \text{ m/s}$$

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$$\vec{p} = 3.6 \text{ kg.m/s}$$

$$\frac{0.2}{0.2} \left(\frac{v}{v} \right)$$

The momentum of law of conservation:-

"The momentum of an isolated system remains constant". In the absence of an external force (an isolated system) the final momentum ' P_f ' of the system must be equal to the initial momentum ' P_i '. If no net force acts on a system of particles, the total momentum of the system cannot change.

Mathematical form :-

$$0 = \frac{\Delta \vec{p}}{\Delta t} \quad \text{or} \quad 0 = \frac{\vec{p}_f - \vec{p}_i}{\Delta t}$$

$$\vec{p}_f = \vec{p}_i \quad \text{or} \quad m_f \vec{v}_f = m_i \vec{v}_i$$

Condition :-

for isolated system, there is no net force acting $\vec{F} = 0$ therefore Newton's second law in term of momentum.

$$\frac{Q: 2(vi)(or)}{Q: 2(vi)(or)}$$

① Principle of moments:-

Second condition of equilibrium is also called principle of moments that state:-

"for an object in equilibrium, the sum of clockwise moments taken about the pivot must be the equal to the sum of the anticlockwise moments taken about the same pivot."

② Data:-

$$d_1 = 0.6 \text{ m}$$

$$W_1 = 250 \text{ N}$$

$$d_2 = ?$$

$$W_2 = 200 \text{ N}$$

Formula:-

$$d_1 \times W_1 = d_2 \times W_2$$

$$= 0.6 \text{ m} \times 250 \text{ N} = 200 \text{ N} \times d_2$$

$$= \frac{0.6 \text{ m} \times 250 \text{ N}}{200 \text{ N}} = d_2$$

$$= \frac{150 \text{ m}}{200 \text{ N}} = d_2$$

$$= 0.75 \text{ Nm} = d_2$$

Q: 2 (vii)

Q: 2 (vii)

~~first~~ ~~cond~~ ~~eq~~ condition of equilibrium:-

Sum of all the vectors acting on the forces acting on the body must be zero. then the first condition of equilibrium is satisfied. Mathematically, the sum of all the forces is

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 \dots = 0$$

$$\sum \vec{F} = 0$$

Second condition condition of equilibrium:-

Sum of all the vectors of all torques acting on the body must be zero then the second condition of equilibrium is satisfied. Mathematically,

$$\vec{\tau}_{net} = \tau_1 + \tau_2 + \tau_3 \dots = 0$$

$$\sum \tau = 0$$

Q: 2 (viii)

Q: 2 (viii)

Three methods of ~~frictions~~ reducing frictions:-

* polishing:-

Polishing can make the surface smoother that can reduce frictions.

* By using ball bearings :-

This method converts the sliding friction into rolling friction by use of ball bearings.

* By applying Lubricants (oil or Grease) to surfaces :-

Friction of certain liquid is less than that of solid surfaces, therefore, oil or grease is applied between the machinery items.

Q:2 (ix)

Terminal velocity :-

The constant maximum velocity that is attained and maintained by an object while falling through air (or any other resistive medium) is called terminal velocity.

For humans, terminal speed in air Skydivers change their effective size and orientation by bending, twisting and stretching their body parts. This allows them to manipulate drag and thereby

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allowing them to control speed and direction of motion.

$$\frac{Q.2(x)(or)}{Q.2(x)(or)}$$

Orbital speed :-

Orbital speed is the constant speed at a satellite (or planet) moves along its circular orbit around a central body. It is the speed required to maintain a stable orbit at a given distance from the center of the gravitating body.

Formula:- $v = 2\pi r / T$

When v = Orbital time speed (m/s)
 r = radius of orbit (m), T = orbital period (time for one complete revolution in seconds), and $\pi \approx 3.14$.

The circumference of the orbit is $2\pi r$ and speed = s/t .

Why same orbital speed :- The orbital speed of a satellite depends only on the radius of the orbit and the mass of the central body (Earth) Not on the mass or size of the satellite itself. For a given orbital distance, the required centripetal force (F_c) is

is provided by gravity ($F_g = GMm/r^2$)
 Setting them equal,

$$\frac{mv^2}{r} = \frac{GMm}{r^2}$$

$$v^2 = \frac{GM}{r}$$

Therefore, all satellites at the same
 at distance orbit at the same
 speed regardless of their mass.

$$\frac{Q_1 Q_2}{r^2} = \frac{Q_1 Q_2}{r^2}$$

Physics:

Physics is the branch of science
 that describes matter, energy, space,
 time, their mutual connections and
 interaction throughout the universe.

For bran. 4 branches of physics:

* **Classical physics:** Classical is the
 branch of physics in which we study
 the motion of ~~mas~~ ~~macro~~ macroscopic
 objects under the action of forces.

* **Climate physics:** Climate physics is
 the branch of physics that deals with
 the study of the Earth's climate.

Particle physics :- Particle physics or (high energy physics) is the study of fundamental particles and forces between electrically charged particles.

Thermal physics :-

Thermal physics is the collective study of statistical mechanics, thermodynamics and kinetic theory of gases.

Section = C

Q.3

Q.3

First law of Newton :-

"If the net external force acting on an object is zero, the object will maintain its state of rest or uniform motion (constant velocity)."

Mathematically,

$$\vec{F}_{\text{net}} = 0$$

$$\Delta \vec{v} = 0 \quad \text{or} \quad \vec{a} = 0$$

For example :-

When a soccer ball is kicked, it experiences the unbalanced force. The ball keeps rolling on the ground until another unbalanced

force alters its movement.

* Newton's second law of motion:

"The acceleration of an object as produced by a net force is directly proportional to the magnitude of the net force, and inversely proportional to the mass of the object."

Mathematically,

$$\vec{F}_{\text{net}} = m\vec{a}$$

For Example:-

When the same force (3000N) acts on a bus (12,000Kg) it accelerates at only 0.25 m/s^2 . But a car (1200Kg) accelerate accelerates at 2.5 m/s^2 under the same force.

So the conclusion is :-

"less mass means greater acceleration."

* Third Law of Newtons :-

"Whenever one object exerts a force on a second object, the second object exerts an equal and opposite force on the first object."

Mathematically:

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

For Example:-

When we punch an object or something as an action, we also get a force onto our hands and legs as a reaction force. That is why we feel pain when punching a wall, or falling on the ground.

Limitations of Newton's Law of motion:-

- Newton's law are not ~~separately~~ readily applied on the very small scale: As one goes to extremely low energies on the atomic scale, position and acceleration are not well defined, where the concepts of quantum mechanics takes over.
- Newton's law are not applied for objects moving at high speed (speeds close to the speed of light) relativistic effects complicate the dynamics at high speeds and high energy energies. In such situations we would require to use relativistic mechanics.

0.4
0.4

Momentum:-

The product of mass (m) and velocity ($\vec{v}$) is called momentum (denoted by P or $\vec{p}$). Mathematically

$$\vec{p} = m\vec{v}.$$

Momentum is a vector quantity that also has direction same with the velocity. and have SI unit ($\text{kg}\cdot\text{m/s}$ or Ns).

Impulse:-

Impulse is the product of the force exerted on an object and time interval over which the force acts and is often given the symbol ' J '. Impulse is a vector quantity and the direction of the impulse is the same direction of the force that causes it and have the same SI unit of momentum.

Derivation:-

From newton's second law:-

$$F_{\text{net}} = ma$$

Since acceleration:-

$$a = \frac{v_f - v_i}{t}$$

Substituting (2) into (1) :-

$$F_{\text{net}} = \frac{m(v_f - v_i)}{t}$$

multiplying both sides by Δt

$$F_{\text{net}} \times \Delta t = mv_f - mv_i$$

$$= F_{\text{net}} \Delta t = p_f - p_i$$

$$= F_{\text{net}} \Delta t = \Delta \vec{p}$$

Impulse = Change in ~~mass~~ momentum :-

$$F_{\text{net}} = \frac{\Delta p}{\Delta t}$$

b) i) Data :-

$$m = 1200 \text{ kg}$$

$$v_i = 8 \text{ m/s}$$

$$v_f = 0 \text{ m/s}$$

$$\Delta t = 55$$

$$\text{Formula :- } F = \frac{(mv_f - mv_i)}{\Delta t}$$

$$F = \frac{(1200 \times 0) - 1200 \times 8}{5}$$

$$F = \frac{0 - 9600}{5}$$

$$F = \frac{-9600}{5}$$

$$F = -1920$$

bii) Data:

$$m = 10,000 \text{ kg}$$

$$v_i = 8 \text{ m/s}$$

$$v_f = 0 \text{ m/s}$$

$$\Delta t = 5 \text{ s}$$

Formula:

$$F_{\text{net}} = \frac{mv_f - mv_i}{\Delta t}$$

$$= \frac{(10,000 \times 0 \text{ m/s} - 10,000 \times 8 \text{ m/s})}{5 \text{ s}}$$

$$= \frac{10,000 \text{ m/s kg} \cdot \text{m/s} - 80,000 \text{ kg} \cdot \text{m/s}}{5 \text{ s}}$$

$$= \frac{(0 \text{ m/s} - 80,000 \text{ kg} \cdot \text{m/s})}{5 \text{ s}}$$

$$= \frac{-80,000 \text{ kg} \cdot \text{m/s}}{5 \text{ s}}$$

$$= -16,000 \text{ N}$$

_____ Q. 5 _____
 _____ A: 5 _____



Equilibrium:-

Equilibrium is the state in which all the individual forces and moments of forces or torques exerted upon an object are balanced. That means that net force and torques on the object are zero. The object is said to be in equilibrium.



First condition of equilibrium:-

Sum When the vector sums of all forces acting on a body is zero then the first condition is satisfied. Mathematically if F_{net} is the sum of forces $F_1 + F_2 + F_3 \dots F_n$ then

$$\vec{F}_{net} = F_1 + F_2 + F_3 \dots + F_n = 0$$

$$\sum \vec{F}_{net} = 0$$



Second condition of equilibrium:-

When the vector sum of all torques acting on a body is zero then the second condition of equilibrium is satisfied. Mathematically if τ_{net} is the sum of torques $\tau_1 + \tau_2 + \tau_3 \dots \tau_n$ then,

$$\tau_{\text{net}} = \tau_1 + \tau_2 + \tau_3 + \dots + \tau_n = 0$$

$$\sum \tau = 0$$

Principle of moments -

Second condition of equilibrium is also known as principle of moments, which states that;

"For an object in equilibrium, the sum of all the clockwise moment must be taken about the pivot must be equal to the sum of the anti-clockwise moment must be taken about the same pivot."

Data :-

$$F_1 = 4\text{ N}$$

$$D_1 = 0.2\text{ m}$$

$$F_2 = 2\text{ N}$$

$$D_2 = 0.4\text{ m}$$

$$\tau_{\text{net}} = \tau_{D_1 \times F_1} = -D_2 \times F_2$$

$$0.2\text{ m} \times 4\text{ N} = -0.4\text{ m} \times 2\text{ N}$$

$$0.8\text{ Nm} = -0.8\text{ Nm}$$

the net force torque is zero

♥ Types of equilibrium and role of CG (centre of gravity):-

CG is the point where the entire weight of a body appears to act. Its behaviour when a body is displaced determines this type of equilibrium.

* Stable equilibrium:-

A body is in stable equilibrium if when slightly tilt and after realizing it returns to its original position.

* It's centre of mass rises when it is displaced. It regains its position back because its weight has a moment of force about the point of contact that acts to reduce the displacement.

FOR EXAMPLE:-

Consider a book lying on the table. Tilt the book slightly about its one edge by lifting it from the opposite side. It returns to its previous position where sets free.

Unstable equilibrium:-

A body is in unstable equilibrium if it moves slightly further away when it is displaced because there is movement which increases the displacement.

* A body is in unstable equilibrium if it's move further away from its previous position when slightly displaced and replaced.

* It is the centre of mass falls which when slightly displaced because there is a moment which increases the displacement.

Neutral equilibrium:-

A body is in neutral equilibrium if it stays in its new position when displaced.

* Its centre of mass does not rise or fall because there is no moment to increase or decrease the displacement.

_____ Q:6 _____
 _____ Q:6 _____

a) Centripetal force:-

The force that pulls an object out of its straight line path and into a circular path is called centripetal (center seeking) force.

Mathematically,

$$F_c = \frac{mv^2}{r}$$

For example:-

A satellite orbiting Earth follows a circular path. The gravitational pull of the Earth provides the centripetal force needed to keep the satellite in orbit. The gravitational force ($F = GMm/r^2$) act toward the center of the Earth, continuously pulling the satellite inward and preventing it from flying off in a straight line.

b) Data:-

mass = 5 kg

$r = 1.5 \text{ m}$

$v = 25 \text{ m/s}$

Formula: mv^2/r

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$$F_c = \frac{mv^2}{r}$$

$$= F_c = \frac{5 \times (25)^2}{1.5}$$

$$F_c = \frac{5 \times 625}{1.5}$$

$$F_c = \frac{3125}{1.5}$$

$$F_c = 2083.3 \text{ N}$$

c) Earth Orbital speed:-

Data:-

Time = 365.25 day

$r = 150 \text{ million Km}$

$$= 150 \times 10^6 \text{ Km}$$

$$= 150 \times 10^8 \text{ Km}$$

$$= 1.50 \times 10^{10} \text{ Km}$$

$$\text{Formula:- } v = \frac{2\pi r}{T}$$

Convert T to second

$$= 365.25 \times 24 \times 60 \times 60$$

$$= 365.25 \times 86400$$

$$3.156 \times 10^7 \text{ s}$$

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$$v = \frac{2\pi r}{T}$$

$$v = \frac{2(3.14) \times 1.5 \times 10^{11}}{3.156 \times 10^7}$$

$$= \frac{9.42 \times 10^{11}}{3.156 \times 10^7}$$

$$= 2.98 \times 10^4 \text{ m/s}$$

$$v = 29.8 \text{ km/s} \approx 29,800 \text{ m/s}$$

Converting into R/hr

$$29.8 \times 3600$$

$$= 107,280 \text{ km/hr}$$